Asset Financing Restraint and the Cross Section of Stock Returns

I. M. Harking

October 21, 2025

Abstract

This paper studies the asset pricing implications of Asset Financing Restraint (AFR), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on AFR achieves an annualized gross (net) Sharpe ratio of 0.52 (0.40), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 22 (16) bps/month with a t-statistic of 3.05 (2.32), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Net debt financing, Net external financing, Change in financial liabilities, Asset growth, Inventory Growth, change in ppe and inv/assets) is 18 bps/month with a t-statistic of 2.57.

1 Introduction

Financial markets efficiently incorporate information into asset prices when firms face production decisions under uncertainty. The cross-section of expected returns reflects heterogeneous exposure to systematic risks arising from firms' real investment decisions and their ability to adjust production in response to economic shocks. Despite extensive research on the production-based foundations of asset pricing, we lack complete understanding of how firms' financing constraints for productive assets interact with their investment decisions to generate cross-sectional variation in risk premia. This paper examines Asset Financing Restraint (AFR), a measure that captures firms' conservative use of debt financing relative to their productive asset base, as a predictor of stock returns. We demonstrate that AFR proxies for firms' exposure to investment frictions and adjustment costs in a production economy, generating systematic variation in conditional risk premia that existing factor models fail to fully explain.

In a production-based asset pricing framework, firms that exhibit high asset financing restraint maintain lower leverage relative to their productive assets, creating a buffer that reduces adjustment costs when investment opportunities arise Zhang (2005). These firms face lower marginal costs of adjusting capital stock in response to productivity shocks, as they can more readily access debt markets without approaching binding financing constraints Livdan et al. (2009). The q-theory of investment predicts that firms with lower adjustment costs earn lower expected returns because they can more flexibly respond to negative productivity shocks, reducing their systematic risk exposure Cochrane (1991). Conversely, firms with low asset financing restraint operate closer to their debt capacity, facing higher marginal costs when adjusting investment. These binding constraints amplify the firms' exposure to aggregate productivity shocks, as they cannot optimally smooth investment over the business cycle Gomes and Schmid (2010). The inability to adjust costlessly transforms temporary productivity shocks into persistent effects on firm value, increasing systematic risk.

The production-based interpretation of AFR connects directly to the investment-based factor models that price the cross-section of returns. High AFR firms exhibit characteristics similar to conservative investment firms, which Cooper et al. (2008) show earn lower returns due to decreasing returns to scale in production. The restraint in asset financing reflects optimal investment policy when firms face convex adjustment costs, leading to time-varying risk premia that correlate with the investment and profitability factors documented by Fama and French (2015). Furthermore, AFR captures variation in firms' distance to default and financial flexibility, characteristics that Whited and Wu (2006) demonstrate affect firms' stochastic discount factors through their impact on the shadow cost of external financing.

The mechanism linking AFR to expected returns operates through the interaction between operating leverage and financial constraints in production economies. Firms with high AFR maintain operational flexibility that allows them to exploit positive productivity shocks without facing steep increases in the marginal cost of external finance Carlson et al. (2004). This flexibility is particularly valuable during periods of high aggregate uncertainty, when the option value of waiting to invest increases. The cross-sectional variation in AFR thus captures differences in firms' exposure to systematic risks arising from time-varying investment opportunities and the real options embedded in production decisions Berk and Green (2004). The production-based framework predicts that AFR should negatively predict returns, as firms maintaining higher financing restraint face lower costs of adjusting production and consequently lower compensation for bearing systematic risk.

We find strong evidence that Asset Financing Restraint predicts cross-sectional variation in stock returns. A value-weighted long-short portfolio that buys stocks with high AFR and sells stocks with low AFR generates an average monthly return of 26 basis points with a t-statistic of 3.73, corresponding to an annualized Sharpe ratio of 0.52. The strategy's alpha remains economically and statistically significant across standard factor models, with the monthly alpha relative to the Fama-French six-factor model equaling 22 basis points with a t-statistic of 3.05. The long-short strategy loads significantly on the CMA (investment) factor with a coefficient of 0.28 and t-statistic of 5.87, consistent with the production-based interpretation that AFR captures variation in firms' investment frictions and adjustment costs.

The predictive power of AFR remains robust across different portfolio construction methodologies and is not confined to small stocks. Among the largest quintile of stocks by market capitalization, the AFR strategy generates an average monthly return of 36 basis points with a t-statistic of 4.09, and alphas ranging from 28 to 40 basis points per month relative to standard factor models. After accounting for transaction costs using the composite bid-ask spread measure, the value-weighted AFR strategy constructed with NYSE breakpoints delivers a net monthly return of 20 basis points with a t-statistic of 2.88, and a net six-factor alpha of 16 basis points with a t-statistic of 2.32. The strategy significantly expands the mean-variance efficient frontier even after accounting for trading costs, with positive net generalized alphas relative to all five factor models tested.

The economic significance of AFR persists after controlling for related anomalies from the factor zoo. When we control for the six most closely related predictors—Net debt financing, Net external financing, Change in financial liabilities, Asset growth, Inventory growth, and Change in PPE and inventory/assets—the AFR strategy maintains an alpha of 18 basis points per month with a t-statistic of 2.57. The gross Sharpe ratio of 0.52 places AFR in the 91st percentile of 212 anomalies tested, while the net Sharpe ratio of 0.40 ranks in the 95th percentile after accounting for implementation costs. These results demonstrate that AFR captures a distinct source of systematic risk related to firms' production decisions and financing constraints that existing factors and anomalies do not fully explain.

This paper contributes to the production-based asset pricing literature by identifying Asset Financing Restraint as a robust predictor of cross-sectional returns that captures firms' exposure to investment frictions and adjustment costs. While Cooper et al. (2008) document that asset growth negatively predicts returns and Lyandros and Sun (2016) show that cash-based operating profitability explains cross-sectional variation, our AFR measure provides a direct link between firms' financing decisions for productive assets and their systematic risk exposure. Unlike the broad asset growth anomaly, AFR specifically captures the interaction between leverage constraints and investment opportunities in a production economy. Our findings extend Livdan et al. (2009), who develop theoretical predictions about financial frictions and expected returns, by providing empirical evidence that firms' restraint in financing productive assets generates systematic variation in risk premia.

Methodologically, we employ the comprehensive testing framework of Novy-Marx and Velikov (2023) to establish the robustness of AFR across multiple dimensions, including alternative portfolio constructions, transaction costs, and controls for related anomalies. This rigorous approach addresses concerns raised by Hou et al. (2015) about data mining and McLean and Pontiff (2016) about the deterioration of anomaly profits post-publication. Our analysis demonstrates that AFR maintains significant predictive power even after controlling for the investment and profitability factors in the Fama and French (2015) five-factor model, suggesting that existing production-based factors do not fully capture the information in firms' asset financing decisions. The robustness of AFR to transaction costs and its strong performance among large-cap stocks distinguish it from many anomalies that derive their power primarily from small, illiquid stocks.

The broader implications of our findings extend to both asset pricing theory and investment practice. From a theoretical perspective, AFR provides empirical support for production-based models that emphasize the role of financial frictions and adjustment costs in generating cross-sectional variation in expected returns. The measure's significant loading on the CMA factor, combined with its incremental predictive power, suggests that firms' financing restraint captures a distinct dimension of investment-related risk. For practitioners, the AFR strategy offers an implementable investment approach that generates significant risk-adjusted returns even after accounting for trading costs. The strategy's performance among large-cap stocks and its robustness to various portfolio construction methods indicate that it represents a genuine source of alpha rather than a statistical artifact. Our results suggest that incorporating measures of firms' financing restraint for productive assets can improve both academic models of cross-sectional returns and practical portfolio construction.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data for the relevant variables, spanning several decades of financial reporting. We focus on industrial firms and exclude financial institutions and utilities due to their distinct regulatory environments and capital structures.Asset Financing Restraint signal relies on two key accounting variables from firms' financial statements. Long-term debt issuance (DLTIS) represents the cash proceeds from the issuance of long-term debt during the fiscal year, capturing new borrowing activities that extend the firm's debt maturity profile. Property, plant, and equipment gross (PPEGT) measures the total historical cost of tangible fixed assets before accumulated depreciation, representing the firm's cumulative investment in productive capacity and infrastructure. These variables provide insight into firms' financing activities and their existing asset base, respectively.construct the

Asset Financing Restraint signal by calculating the year-over-year change in long-term debt issuance scaled by lagged gross property, plant, and equipment: (DLTIS(t) - DLTIS(t-1)) / PPEGT(t-1). This measure captures the change in a firm's long-term borrowing activity relative to its existing tangible asset base, providing a normalized metric of financing capacity utilization. A higher value indicates an acceleration in long-term debt financing relative to the firm's physical capital stock, while a lower or negative value suggests a deceleration or reduction in such financing activities. We compute this signal using annual observations based on fiscal year-end values and require non-missing values for both DLTIS and PPEGT in consecutive years. To ensure data quality, we winsorize the signal at the 1st and 99th percentiles to mitigate the influence of extreme outliers.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the AFR signal. Panel A plots the time-series of the mean, median, and interquartile range for AFR. On average, the cross-sectional mean (median) AFR is -0.47 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input AFR data. The signal's interquartile range spans -0.18 to 0.18. Panel B of Figure 1 plots the time-series of the coverage of the AFR signal for the CRSP universe. On average, the AFR signal is available for 5.83% of CRSP names, which on average make up 7.10% of total market capitalization.

4 Does AFR predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on AFR using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high AFR portfolio and sells the low AFR portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short AFR strategy earns an average return of 0.26% per month with a t-statistic of 3.73. The annualized Sharpe ratio of the strategy is 0.52. The alphas range from 0.22% to 0.31% per month and have t-statistics exceeding 3.05 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.28, with a t-statistic of 5.87 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 474 stocks and an average market capitalization of at least $1,675 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 18 bps/month with a t-statistics of 4.02. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for twenty-three exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between -8-26bps/month. The lowest return, (-8 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -1.40. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the AFR trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in eighteen cases.

Table 3 provides direct tests for the role size plays in the AFR strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and AFR, as well as average returns and alphas for long/short trading AFR strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the AFR strategy achieves an average return of 36 bps/month with a t-statistic of 4.09. Among these large cap stocks, the alphas for the AFR strategy relative to the five most common factor models range from 28 to 40 bps/month with t-statistics between 3.11 and 4.55.

5 How does AFR perform relative to the zoo?

Figure 2 puts the performance of AFR in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the AFR strategy falls in the distribution. The AFR strategy's gross (net) Sharpe ratio of 0.52 (0.40) is greater than 91% (95%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a $1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a $1 invested in the AFR strategy (red line).[2] Ignoring trading costs, a $1 invested in the AFR strategy would have yielded $3.35 which ranks the AFR strategy in the top 7% across the 212 anomalies. Accounting for trading costs, a $1 invested in the AFR strategy would have yielded $2.04 which ranks the AFR strategy in the top 6% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the AFR relative to those. Panel A shows that

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

[2] The figure assumes an initial investment of $1 in T-bills and $1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

the AFR strategy gross alphas fall between the 59 and 70 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The AFR strategy has a positive net generalized alpha for five out of the five factor models. In these cases AFR ranks between the 77 and 83 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does AFR add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of AFR with 209 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price AFR or at least to weaken the power AFR has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of AFR conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{AFR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{AFR} AFR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X

[3] When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{AFR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on AFR. Stocks are finally grouped into five AFR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted AFR trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on AFR and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the AFR signal in these Fama-MacBeth regressions exceed 2.25, with the minimum t-statistic occurring when controlling for Asset growth. Controlling for all six closely related anomalies, the t-statistic on AFR is 1.32.

Similarly, Table 5 reports results from spanning tests that regress returns to the AFR strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the AFR strategy earns alphas that range from 18-20bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.61, which is achieved when controlling for Asset growth. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the AFR trading strategy achieves an alpha of 18bps/month with a t-statistic of 2.57.

7 Does AFR add relative to the whole zoo?

Finally, we can ask how much adding AFR to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the AFR signal.[4] We consider one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 155-anomaly combination strategy grows to $993.77, while $1 investment in the combination strategy that includes AFR grows to $1152.06.

8 Conclusion

This study provides robust evidence that Asset Financing Restraint (AFR) represents a significant and economically meaningful predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that AFR-based trading strategies generate substantial risk-adjusted returns that persist even after accounting for well-established risk factors and closely related anomalies.

The key findings reveal that a value-weighted long/short strategy based on AFR delivers an annualized net Sharpe ratio of 0.40, which remains economically signifi-

[4]We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which AFR is available.

cant after accounting for transaction costs. More importantly, the strategy generates a monthly alpha of 16 basis points (t-statistic = 2.32) relative to the Fama-French five-factor model augmented with momentum, indicating that AFR captures unique information about expected returns not subsumed by conventional risk factors. The persistence of significant alpha (18 bps/month, t-statistic = 2.57) even after controlling for six closely related financing and investment strategies—including net debt financing, net external financing, and asset growth—underscores the distinct nature of the AFR signal.

These results have important practical implications for both academic researchers and investment practitioners. For academics, AFR adds to our understanding of how firms' financing constraints and capital allocation decisions influence expected returns, contributing to the broader literature on the investment-based asset pricing anomalies. For practitioners, the economic magnitude and statistical robustness of the AFR signal, combined with its resilience to transaction costs, suggest potential value in incorporating this measure into quantitative investment strategies.

However, several limitations warrant consideration. First, while our analysis accounts for transaction costs using standard assumptions, actual implementation costs may vary depending on market conditions and portfolio size. Second, the sample period and geographic scope of our analysis may limit the generalizability of findings to other markets or time periods. Third, while we control for numerous related factors, the possibility of omitted risk factors or alternative explanations for the AFR premium cannot be entirely ruled out.

Future research should explore several promising avenues. First, investigating the economic mechanisms underlying the AFR anomaly could provide deeper insights into why markets appear to misprice firms based on their asset financing patterns. Second, examining the international evidence for AFR across different markets and regulatory environments would enhance our understanding of its robustness and universality. Third, exploring the time-variation in the AFR premium and its relationship to macroeconomic conditions could reveal important conditional patterns. Finally, investigating potential interactions between AFR and other corporate characteristics, such as governance quality or information asymmetry, may uncover additional nuances in how financing restraints affect stock returns. Understanding these dynamics will be crucial for both advancing asset pricing theory and improving the practical implementation of AFR-based investment strategies.

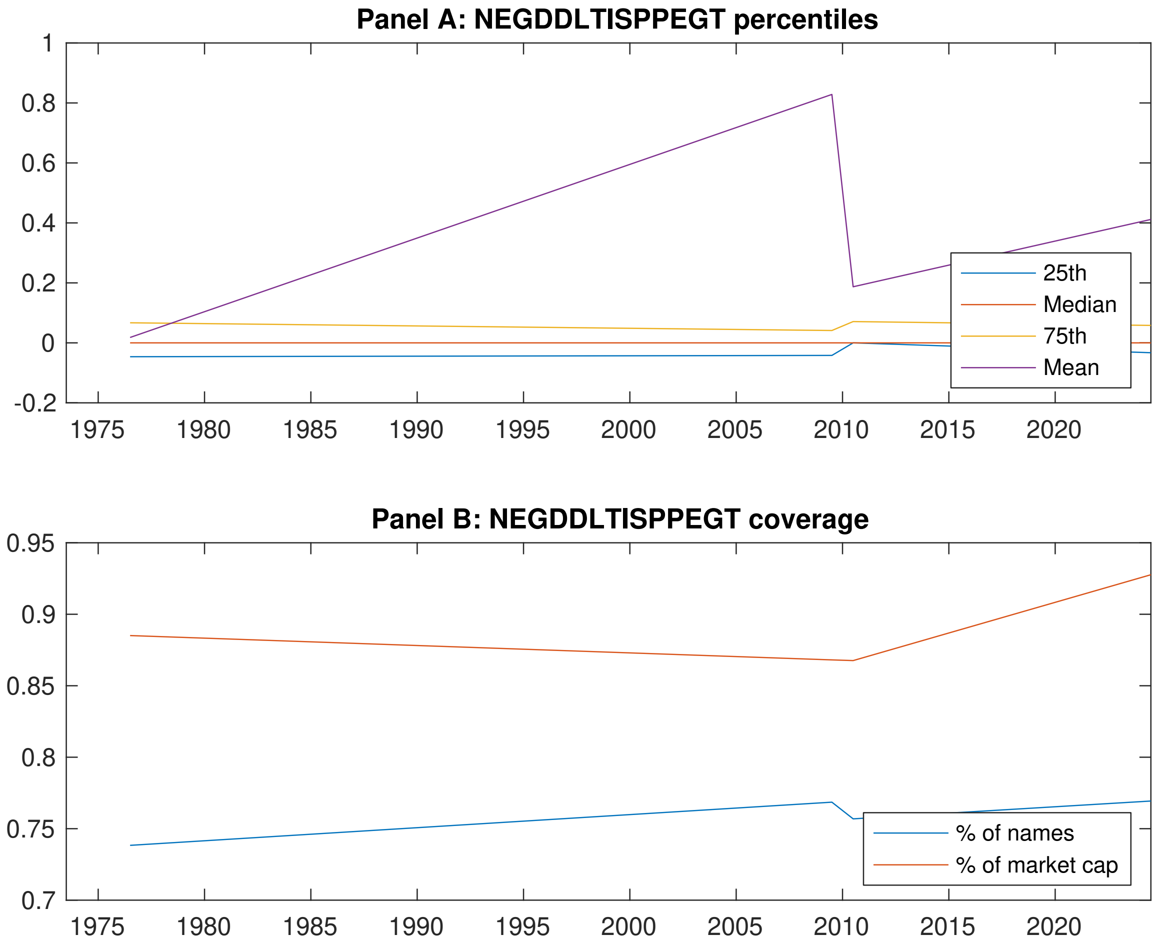


Figure 1: Times series of AFR percentiles and coverage.
This figure plots descriptive statistics for AFR. Panel A shows cross-sectional percentiles of AFR over the sample. Panel B plots the monthly coverage of AFR relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on AFR. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on AFR-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.62	0.66	0.62	0.70	0.88	0.26
	[2.84]	[3.77]	[3.22]	[4.01]	[4.30]	[3.73]
α_{CAPM}	-0.13	0.07	-0.03	0.11	0.18	0.31
	[-2.29]	[1.35]	[-0.46]	[2.21]	[3.43]	[4.44]
α_{FF3}	-0.12	0.05	0.03	0.11	0.20	0.31
	[-2.14]	[1.02]	[0.57]	[2.24]	[3.81]	[4.46]
α_{FF4}	-0.09	0.05	0.08	0.07	0.18	0.27
	[-1.63]	[1.08]	[1.42]	[1.36]	[3.50]	[3.86]
α_{FF5}	-0.09	-0.04	0.06	0.03	0.14	0.24
	[-1.74]	[-0.85]	[1.03]	[0.61]	[2.74]	[3.38]
α_{FF6}	-0.08	-0.03	0.09	0.00	0.14	0.22
	[-1.42]	[-0.63]	[1.67]	[0.05]	[2.62]	[3.05]
Panel B: Fama and French (2018) 6-factor model loadings for AFR-sorted portfolios						
β_{MKT}	1.10	0.97	0.96	0.95	1.05	-0.05
	[86.47]	[86.31]	[73.15]	[82.62]	[86.53]	[-2.80]
β_{SMB}	0.10	-0.11	-0.02	-0.05	0.15	0.05
	[5.18]	[-6.52]	[-0.82]	[-2.66]	[8.16]	[2.05]
β_{HML}	0.00	0.02	-0.16	-0.05	-0.13	-0.13
	[0.00]	[0.99]	[-6.28]	[-2.32]	[-5.56]	[-4.14]
β_{RMW}	0.05	0.15	-0.02	0.09	0.07	0.01
	[2.16]	[6.81]	[-0.73]	[3.91]	[2.88]	[0.47]
β_{CMA}	-0.15	0.15	-0.05	0.18	0.13	0.28
	[-4.06]	[4.67]	[-1.23]	[5.34]	[3.65]	[5.87]
β_{UMD}	-0.03	-0.02	-0.06	0.04	0.01	0.03
	[-2.10]	[-1.47]	[-4.37]	[3.89]	[0.63]	[2.10]
Panel C: Average number of firms (n) and market capitalization (me)						
n	668	474	995	516	644	
me ($\$10^6$)	1675	2634	2004	2612	1678	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the AFR strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas

Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.26	0.31	0.31	0.27	0.24	0.22
			[3.73]	[4.44]	[4.46]	[3.86]	[3.38]	[3.05]
Quintile	NYSE	EW	0.18	0.20	0.19	0.18	0.18	0.18
			[4.02]	[4.65]	[4.35]	[4.16]	[4.06]	[4.00]
Quintile	Name	VW	0.27	0.31	0.32	0.29	0.25	0.24
			[3.86]	[4.40]	[4.49]	[4.09]	[3.57]	[3.41]
Quintile	Cap	VW	0.26	0.31	0.31	0.27	0.23	0.21
			[4.00]	[4.78]	[4.86]	[4.22]	[3.57]	[3.23]
Decile	NYSE	VW	0.33	0.39	0.37	0.34	0.23	0.22
			[3.52]	[4.11]	[3.89]	[3.52]	[2.42]	[2.32]

Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas

Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.20	0.24	0.24	0.22	0.18	0.16
			[2.88]	[3.46]	[3.45]	[3.14]	[2.56]	[2.32]
Quintile	NYSE	EW	-0.08					
			[-1.40]					
Quintile	Name	VW	0.21	0.24	0.24	0.23	0.19	0.18
			[2.98]	[3.40]	[3.44]	[3.25]	[2.71]	[2.55]
Quintile	Cap	VW	0.21	0.25	0.26	0.24	0.19	0.17
			[3.18]	[3.96]	[4.00]	[3.68]	[2.93]	[2.69]
Decile	NYSE	VW	0.26	0.32	0.30	0.28	0.18	0.17
			[2.74]	[3.34]	[3.12]	[2.95]	[1.91]	[1.80]

Table 3: Conditional sort on size and AFR

This table presents results for conditional double sorts on size and AFR. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on AFR. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high AFR and short stocks with low AFR .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results

		AFR Quintiles					AFR Strategies					
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Size quintiles	(1)	0.67	0.87	0.97	0.92	0.84	0.16	0.20	0.17	0.14	0.13	0.11
		[2.28]	[3.29]	[3.50]	[3.40]	[2.78]	[1.61]	[1.97]	[1.69]	[1.30]	[1.25]	[1.04]
	(2)	0.75	0.88	0.85	0.91	0.85	0.10	0.13	0.10	0.12	0.06	0.08
		[2.75]	[3.55]	[3.34]	[3.74]	[3.29]	[1.18]	[1.65]	[1.32]	[1.50]	[0.80]	[1.00]
	(3)	0.83	0.78	0.83	0.80	0.94	0.11	0.16	0.15	0.11	0.16	0.13
		[3.22]	[3.56]	[3.47]	[3.66]	[3.88]	[1.43]	[1.98]	[1.93]	[1.39]	[1.95]	[1.56]
	(4)	0.76	0.81	0.82	0.77	0.87	0.11	0.15	0.13	0.10	0.10	0.08
		[3.25]	[3.85]	[3.71]	[3.77]	[3.92]	[1.49]	[1.94]	[1.66]	[1.25]	[1.31]	[1.04]
	(5)	0.53	0.63	0.58	0.58	0.89	0.36	0.40	0.40	0.34	0.32	0.28
		[2.60]	[3.56]	[3.12]	[3.24]	[4.57]	[4.09]	[4.48]	[4.55]	[3.79]	[3.58]	[3.11]

Panel B: Portfolio average number of firms and market capitalization

		AFR Quintiles					AFR Quintiles				
		Average n					Average market capitalization ($\$10^6$)				
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
Size quintiles	(1)	367	368	369	368	364	33	30	29	29	32
	(2)	101	101	101	102	101	55	56	55	56	55
	(3)	72	73	72	73	73	100	100	96	97	100
	(4)	61	62	62	62	61	219	225	218	222	217
	(5)	57	57	57	57	57	1394	2002	1667	2045	1472

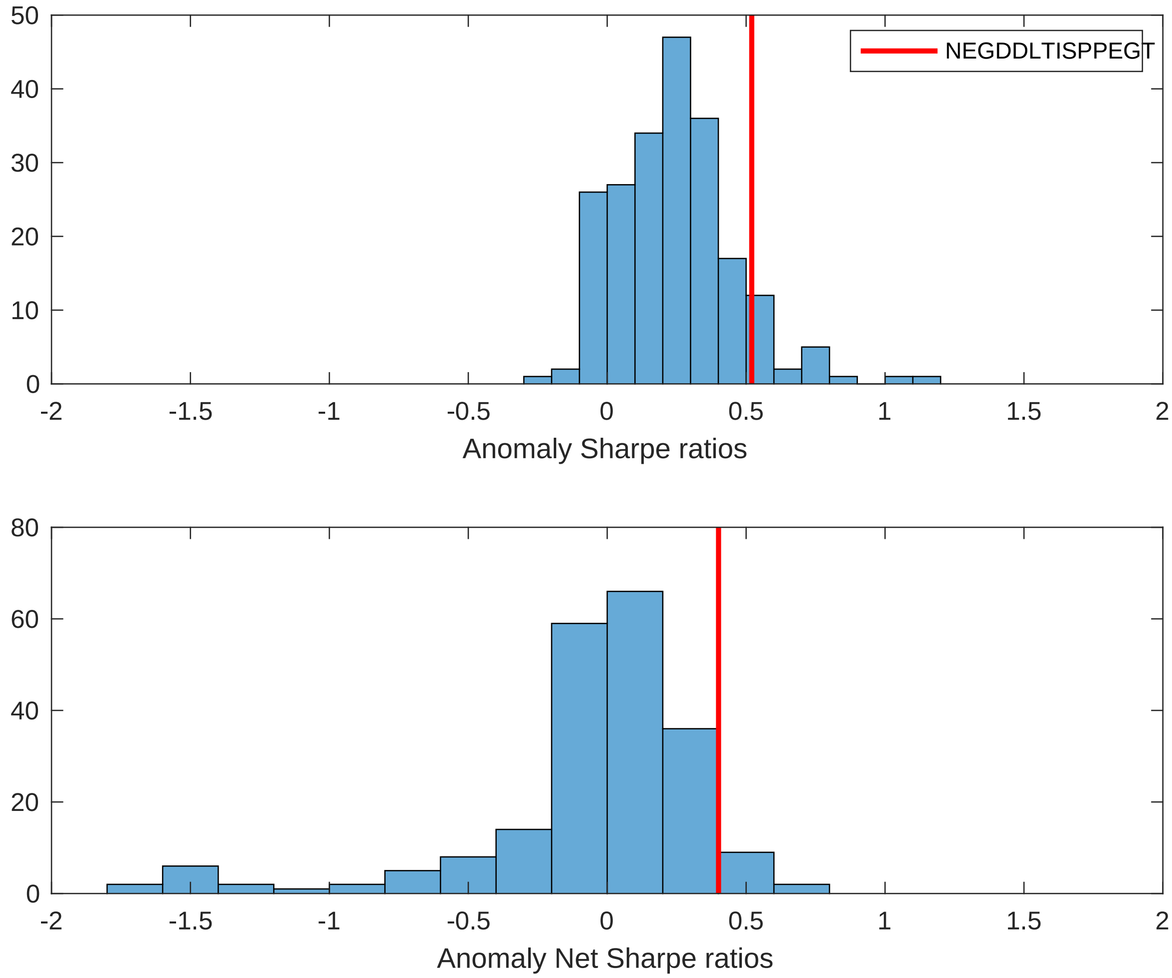


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the AFR with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

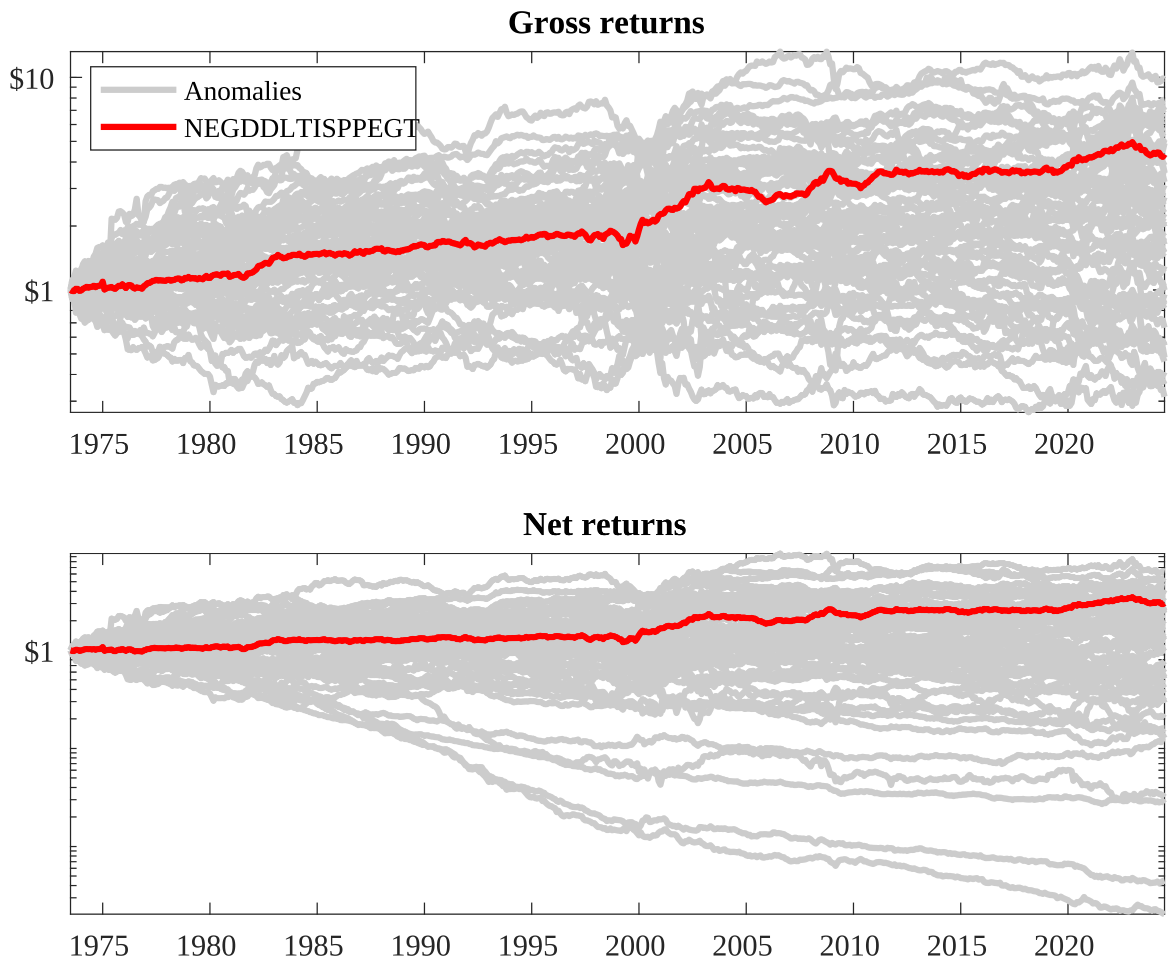
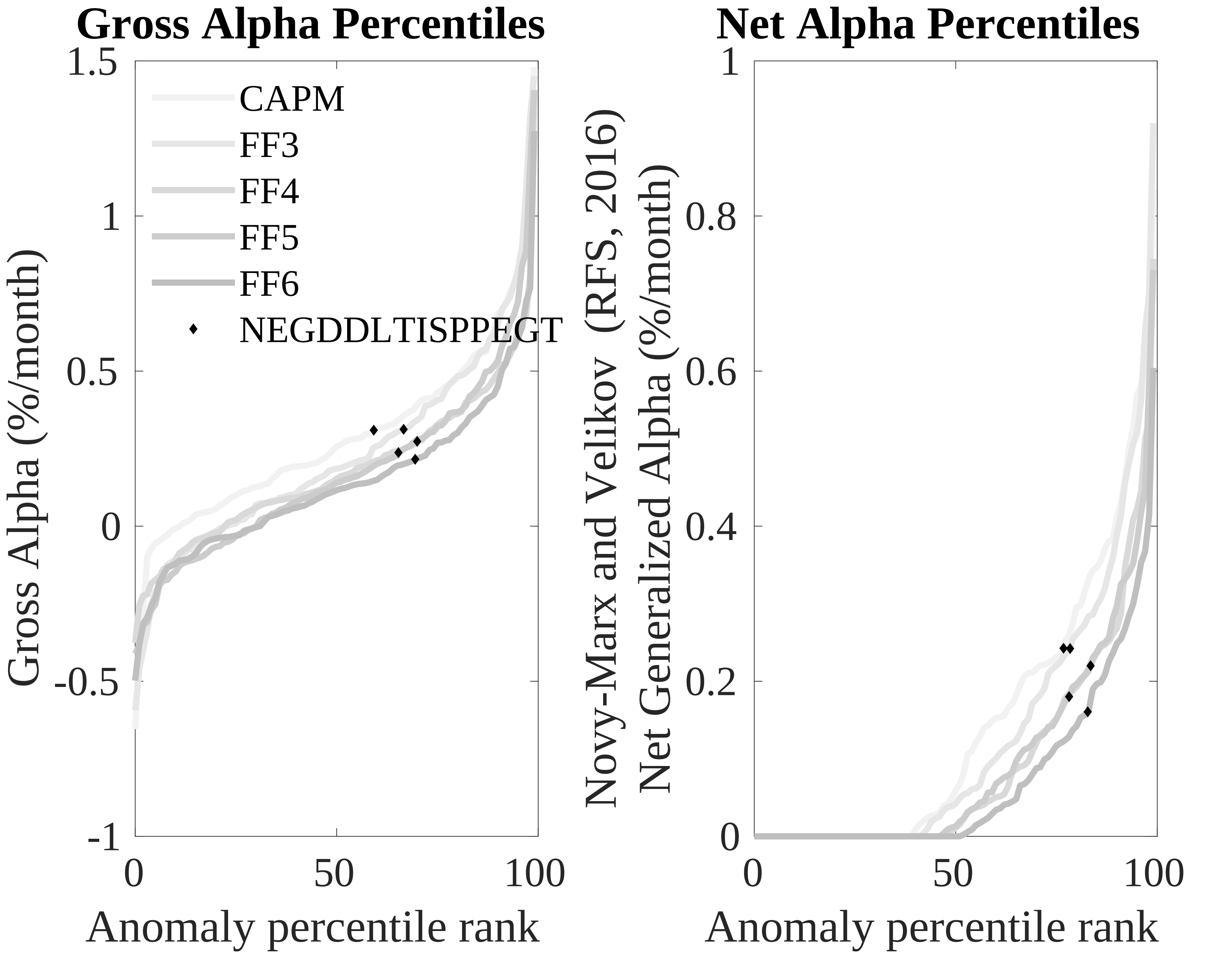


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the AFR trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
Net Alpha Percentiles
CAPM
FF3
FF4
FF5
FF6
NEGDDLTISPPEGT
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
Anomaly percentile rank

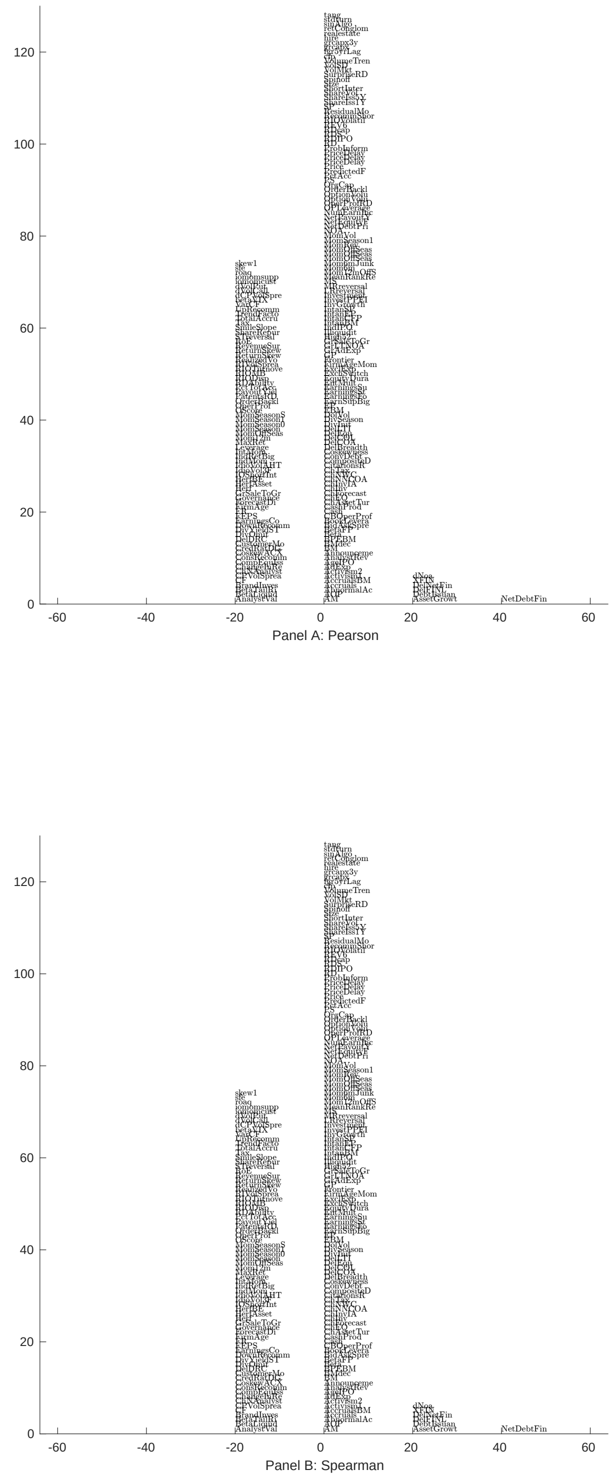


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 209 filtered anomaly signals with AFR. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

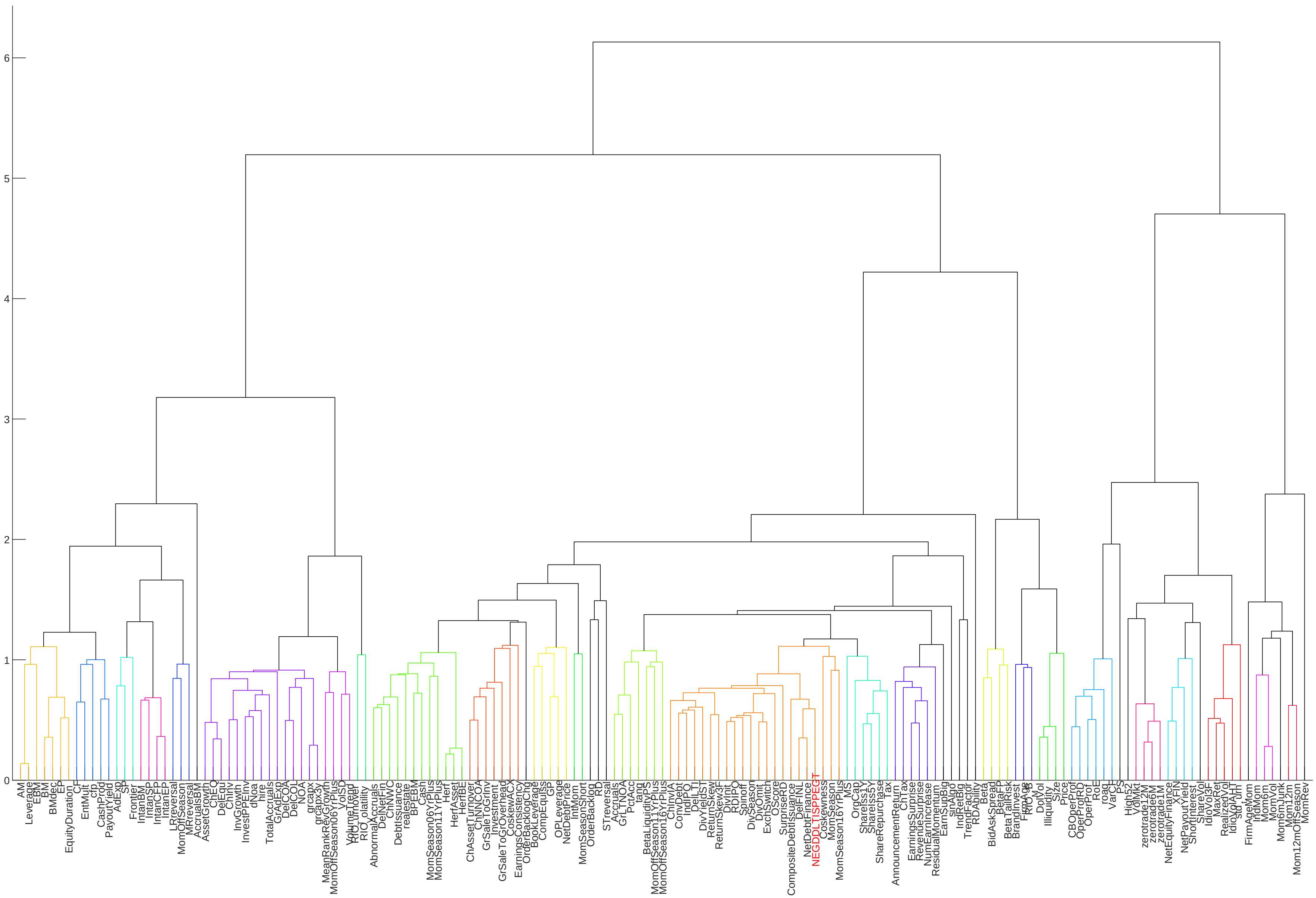


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

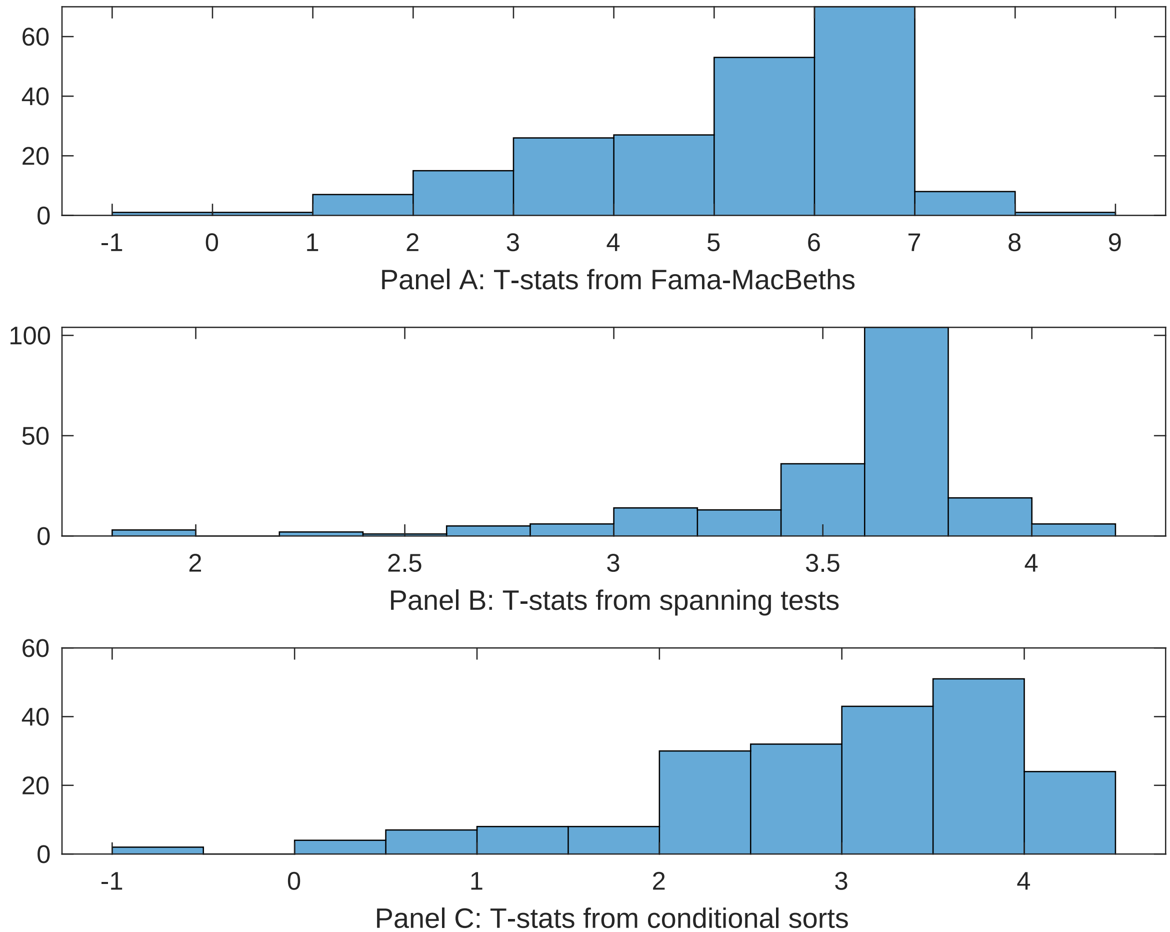


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of AFR conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{AFR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{AFR} AFR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{AFR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on AFR. Stocks are finally grouped into five AFR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted AFR trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on AFR. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{AFR} AFR_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Net debt financing, Net external financing, Change in financial liabilities, Asset growth, Inventory Growth, change in ppe and inv/assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.13	0.14	0.13	0.14	0.14	0.14	0.15
	[5.35]	[5.72]	[5.38]	[5.79]	[5.31]	[5.76]	[5.75]
AFR	0.88	0.10	0.75	0.64	0.15	0.95	0.42
	[3.06]	[4.00]	[2.94]	[2.25]	[5.11]	[3.51]	[1.32]
Anomaly 1	0.19						0.91
	[8.53]						[1.34]
Anomaly 2		0.19					0.11
		[6.30]					[1.82]
Anomaly 3			0.17				-0.79
			[9.16]				[-1.65]
Anomaly 4				0.10			0.51
				[8.27]			[2.35]
Anomaly 5					0.41		0.25
					[6.94]		[0.42]
Anomaly 6						0.17	0.86
						[7.59]	[3.10]
# months	612	612	612	612	612	612	612
$\bar{R}^2$(%)	0	1	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies
This table presents spanning tests results of regressing returns to the AFR trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{AFR} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Net debt financing, Net external financing, Change in financial liabilities, Asset growth, Inventory Growth, change in ppe and inv/assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.19	0.18	0.19	0.20	0.20	0.20	0.18
	[2.73]	[2.61]	[2.74]	[2.85]	[2.93]	[2.89]	[2.57]
Anomaly 1	15.76						11.45
	[4.15]						[2.21]
Anomaly 2		17.80					14.30
		[5.16]					[3.77]
Anomaly 3			9.47				-0.24
			[2.39]				[-0.05]
Anomaly 4				9.79			3.87
				[2.41]			[0.92]
Anomaly 5					4.97		3.92
					[1.82]		[1.37]
Anomaly 6						4.40	-1.83
						[1.39]	[-0.55]
mkt	-4.23	-1.81	-4.04	-4.26	-4.35	-4.32	-2.30
	[-2.65]	[-1.09]	[-2.51]	[-2.64]	[-2.70]	[-2.67]	[-1.38]
smb	4.04	11.15	4.46	4.72	5.98	5.45	9.17
	[1.66]	[4.22]	[1.81]	[1.92]	[2.43]	[2.23]	[3.26]
hml	-11.92	-10.59	-11.92	-13.02	-12.71	-13.02	-10.67
	[-3.87]	[-3.44]	[-3.83]	[-4.18]	[-4.08]	[-4.15]	[-3.42]
rmw	-0.10	-9.01	0.78	1.67	2.35	1.71	-7.61
	[-0.03]	[-2.42]	[0.25]	[0.53]	[0.74]	[0.54]	[-1.99]
cma	24.20	16.69	25.11	16.44	23.86	24.79	9.40
	[5.07]	[3.24]	[5.12]	[2.42]	[4.49]	[4.63]	[1.34]
umd	2.63	3.52	2.83	3.85	3.14	3.64	2.50
	[1.64]	[2.24]	[1.73]	[2.40]	[1.93]	[2.27]	[1.53]
# months	612	612	612	612	612	612	612
$\bar{R}^2$(%)	13	14	11	11	11	10	15

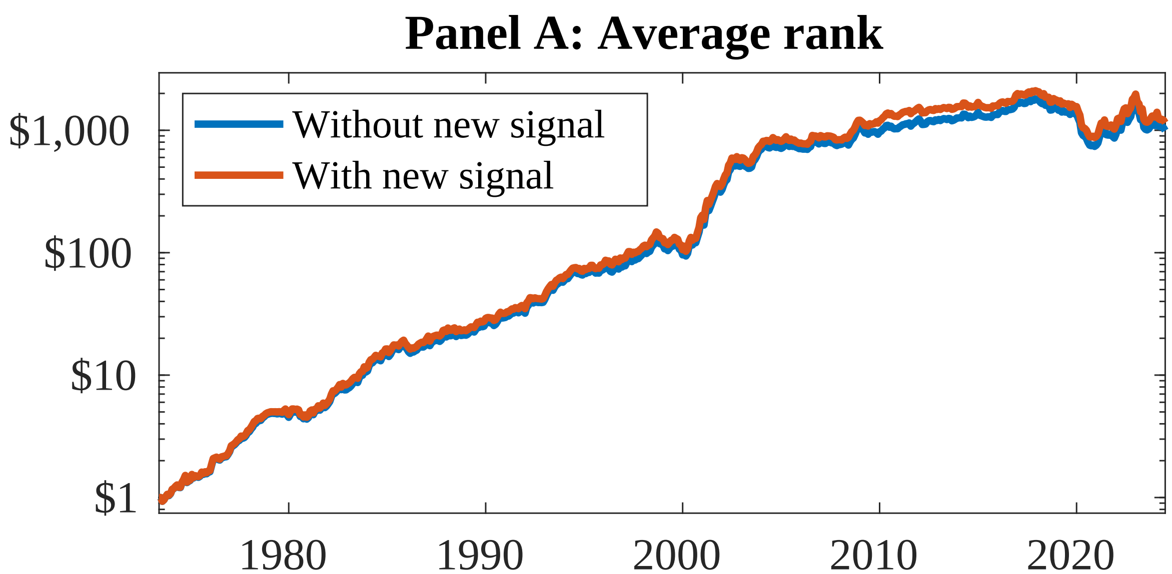


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as AFR. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

References

Berk, J. B. and Green, R. C. (2004). Mutual fund flows and performance in rational markets. *Journal of Political Economy*, 112(6):1269–1295.

Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52:57–82.

Carlson, M., Fisher, A., and Giammarino, R. (2004). Corporate investment and asset price dynamics: Implications for the cross-section of returns. *Journal of Finance*, 59(6):2577–2603.

Chen, A. and Velikov, M. (2022). Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*, Forthcoming.

Chen, A. Y. and Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 27(2):207–264.

Cochrane, J. H. (1991). Production-based asset pricing and the link between stock returns and economic fluctuations. *Journal of Finance*, 46(1):209–237.

Cooper, M. J., Gulen, H., and Schill, M. J. (2008). Asset growth and the cross-section of stock returns. *Journal of Finance*, 63(4):1609–1651.

Detzel, A., Novy-Marx, R., and Velikov, M. (2022). Model comparison with transaction costs. *Journal of Finance, Forthcoming.*

Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.

Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.

Fama, E. F. and French, K. R. (2018). Choosing factors. *Journal of Financial Economics*, 128(2):234–252.

Gomes, J. F. and Schmid, L. (2010). Financing constraints, firm dynamics, and international trade. *Review of Financial Studies*, 23(11):4061–4109.

Hou, K., Xue, C., and Zhang, L. (2015). Digesting anomalies: An investment approach. *Review of Financial Studies*, 28(3):650–705.

Livdan, D., Sapriza, H., and Zhang, L. (2009). Corporate investment and stock returns. *Journal of Financial Economics*, 92(3):412–432.

Lyandros, G. and Sun, L. (2016). Equity valuation and accounting numbers: Applying zhang (2000) to the financial crisis. *Journal of Accounting Research*, 54(2):524–558.

McLean, R. D. and Pontiff, J. (2016). Does academic research destroy stock return predictability? *Journal of Finance*, 71(1):5–32.

Novy-Marx, R. and Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.

Novy-Marx, R. and Velikov, M. (2023). Assaying anomalies. *Working paper*.

Whited, T. M. and Wu, G. (2006). Financial constraints risk. *Review of Financial Studies*, 19(2):531–559.

Zhang, L. (2005). The value premium. *Journal of Finance*, 60(1):67–103.